

Machine Learning Project Report: Student Pass/Fail Prediction

Introduction: This project focuses on building a supervised machine learning system to predict whether a student will pass or fail using academic indicators such as attendance, internal test scores, assignment marks, and study hours.

Objective: To apply and compare multiple classification algorithms to identify the most reliable prediction model.

Methodology:

- Data preprocessing and normalization
- Creation of Pass/Fail target variable
- Train-test splitting (70/30)
- Training and evaluation of multiple ML models

Models Used: Logistic Regression, Decision Tree, KNN, Random Forest

These models represent different learning approaches including linear, rule-based, distance-based, and ensemble learning.

Model Performance/Accuracy:

- Logistic Regression ~ 0.95
- Decision Tree ~ 0.91
- KNN ~ 0.93
- Random Forest ~ 0.94

Evaluation Techniques:

- **Confusion Matrix:** Used to identify correct and incorrect predictions made by the model.
- **Classification Report:** Used to evaluate model performance using precision, recall, and F1-score.
- **Actual vs Predicted Visualization:** Used to visually compare real results with model predictions.
- **Feature Importance Analysis:** Used to identify which features most influence student performance predictions.
- **5-Fold Cross Validation:** Used to check whether the model performs consistently on different data splits.
- **Hyperparameter Tuning (GridSearchCV):** Used to find the best model settings for improved accuracy and performance.

Final Model Selection: The final model was selected based on original accuracy, cross-validation score, and tuned accuracy.

Best Model: Logistic Regression due to its high accuracy, stability, and better generalization ability.

Conclusion: This project successfully demonstrates the use of machine learning for predicting student outcomes. Comparing multiple models, applying cross-validation, and using hyperparameter tuning helped develop a more reliable and accurate prediction system.

GitHub Repo Link: <https://github.com/Sithumini-Anuhansi/student-pass-fail-ml.git>